# 路径故障转移
## ONTAP 9

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# 路径故障转移

## 路径故障转移概述

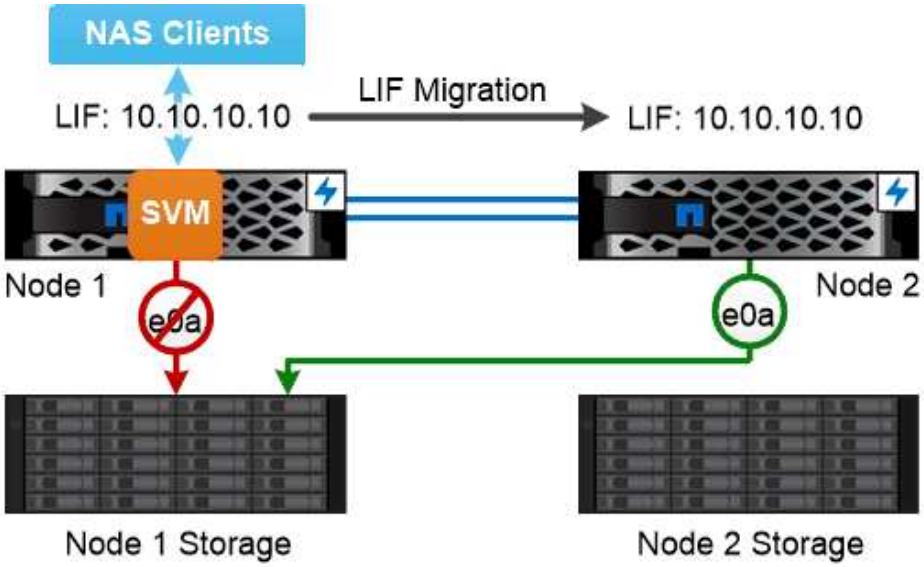
ONTAP 在 NAS 和 SAN 拓扑中管理路径故障转移的方式存在重要差异。链路出现故障后， NAS LIF 会自动迁移到其他网络端口。SAN LIF 不会迁移（除非您在发生故障后手动移动）。相反，主机上的多路径技术会将流量转移到同一 SVM 上的其他 LIF ，但会访问不同的网络端口。

## NAS 路径故障转移

在 NAS LIF 的当前端口出现链路故障后，该 LIF 会自动迁移到正常运行的网络端口。LIF 迁移到的端口必须是 LIF 的 *failover group* 的成员。故障转移组策略 _ 可将数据 LIF 的故障转移目标缩小至数据所属节点及其 HA 配对节点上的端口。

为了便于管理， ONTAP 会为网络架构中的每个 *broadcast domain* 创建一个故障转移组。广播域对属于同一第 2 层网络的端口进行分组。例如，如果您使用 VLAN 按部门（工程，营销，财务等）隔离流量，则每个 VLAN 都会定义一个单独的广播域。每次添加或删除广播域端口时，与广播域关联的故障转移组都会自动更新。

几乎始终最好使用广播域定义故障转移组，以确保故障转移组保持最新。但是，有时可能需要定义与广播域不关联的故障转移组。例如，您可能希望 LIF 仅故障转移到广播域中定义的部分端口中的端口。




*A NAS LIF automatically migrates to a surviving network port after a link failure on its current port.*

- 子网 _*

*subnet* 会在广播域中保留一个 IP 地址块。这些地址属于同一个第 3 层网络，并在创建 LIF 时分配给广播域中的端口。与指定 IP 地址和网络掩码相比，在定义 LIF 地址时指定子网名称通常更简单，更不容易出错。

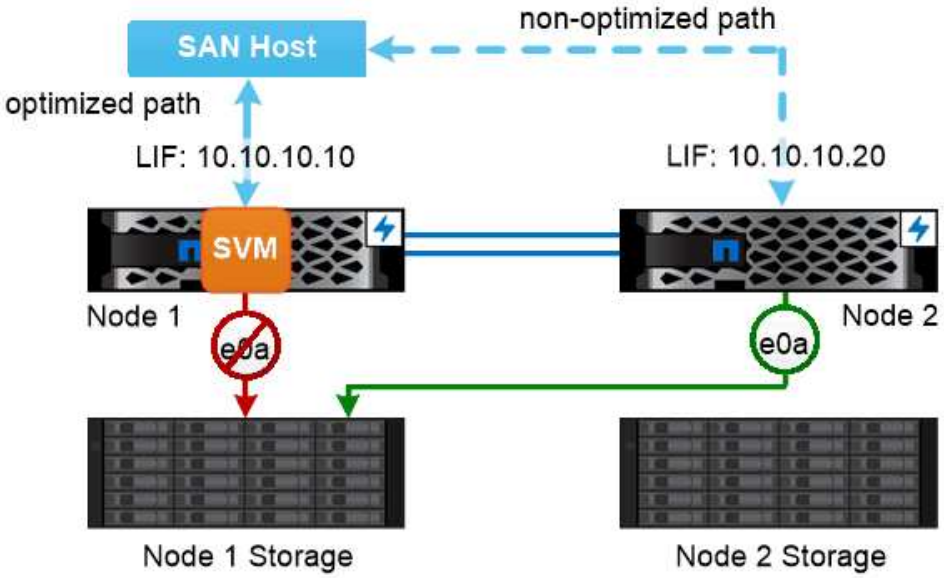
# SAN 路径故障转移

在链路出现故障后， SAN 主机使用 ALUA （非对称逻辑单元访问）和 MPIO （多路径 I/O ）将流量重新路由到运行正常的 LIF 。预定义的路径可确定 SVM 所提供的 LUN 的可能路由。

在 SAN 环境中，主机被视为向 lun *targets* 发出请求的 _*initiator* 。 _ MPIO 可启用从启动程序到目标的多个路径。ALUA 标识最直接的路径，称为 _optim优化 的路径。 _

通常，您可以为 LUN 所属节点上的 LIF 配置多个优化路径，并为其 HA 配对节点上的 LIF 配置多个非优化路径。如果所属节点上的一个端口发生故障，则主机会将流量路由到运行正常的端口。如果所有端口都发生故障，主机将通过非优化路径路由流量。

默认情况下， ONTAP 选择性 LUN 映射（ SLM ）会限制从主机到 LUN 的路径数。新创建的 LUN 只能通过 LUN 所属节点或其 HA 配对节点的路径进行访问。您还可以通过在 *port set* 中为启动程序配置 LIF 来限制对 LUN 的访问。




*A SAN host uses multipathing technology to reroute traffic to a surviving LIF after a link failure.*

在 SAN 环境中移动卷 _*

默认情况下， ONTAP 选择性 LUN 映射（ SLM ） _ 会限制从 SAN 主机到 LUN 的路径数。新创建的 LUN 只能通过 LUN 所属节点或其 HA 配对节点的路径访问，即 LUN 的 *reporting nodes* 。

这意味着，将卷移动到另一个 HA 对上的节点时，您需要将目标 HA 对的报告节点添加到 LUN 映射中。然后，您可以在 MPIO 设置中指定新路径。卷移动完成后，您可以从映射中删除源 HA 对的报告节点。

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